

CPH 200C PS3 · CLINICAL AI BY REGION

Three Governing Systems for Clinical AI

Why deployed clinical AI looks different across the US, Europe, and Asia: policy, funding, technical developments, and landscape

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CPH 200C · Computational Precision Health

2026-06-04



The lens: deployed clinical AI

Three regions, three governing systems for AI

UNITED STATES

Market-led

Light federal + state patchwork; deregulate to 'win the race'. Closed frontier labs, deepest capital.

EUROPE / EU

Comprehensive

Horizontal risk-tiered AI Act (now softening). Sovereignty money, little frontier building.

ASIA

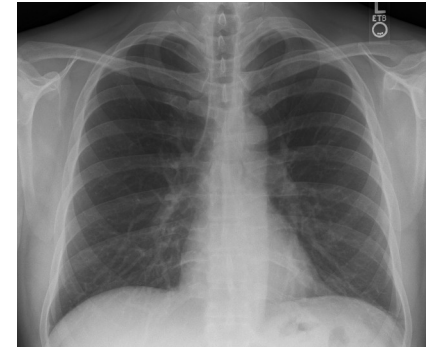
China-led

State-driven core (China: open-weight, population scale). Korea/Japan/India build their own models. Asia owns the AI hardware: TSMC, HBM, Foxconn.

Three governing systems (US / EU / Asia), tested on one case.

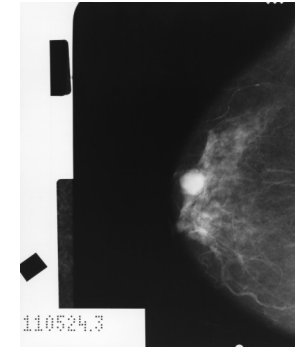
Evidence base: npj Digital Medicine - Nature Medicine - Lancet Digital Health
- JAMA - JMIR - Stanford AI Index 2026 - MIT NANDA

Deployed clinical AI is mostly radiology - a named, in-production system per modality:



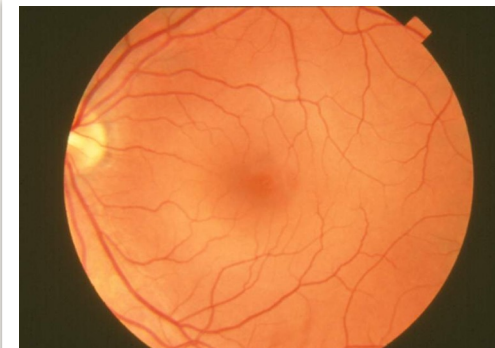
CHEST X-RAY

Oxipit ChestLink (EU)



MAMMOGRAPHY

Lunit INSIGHT (KR)



RETINAL FUNDUS

IDx-DR (US, autonomous)



CHEST CT

Infervision (CN)

Images: public domain / CC0, Wikimedia Commons.

What leaders predicted, and what's different now

CHINA

Kai-Fu Lee

AI Superpowers (2018)

The age of implementation: execution and data, not breakthroughs, decide the race, favoring China.

UNITED STATES

Eric Schmidt et al.

The Age of AI (2021); NSCAI

AI is a national-security race the US must win; warned of China closing the gap fast.

EUROPE / EU

Mario Draghi

EU Competitiveness Report (2024)

Europe regulates but does not build; structurally behind on compute, capital, and scale.

WHAT'S DIFFERENT NOW (2025):

None predicted a Chinese OPEN model (DeepSeek) would reset the frontier while US labs re-closed. Lee's implementation thesis holds, but the ROI gap shows capability is not yet value.

Asia beyond China

INDIA

POPULATION-SCALE

Sarvam (govt-backed sovereign LLM) + Krutrim, under the national IndiaAI Mission. AI layered on UPI / Aadhaar at population scale; Qure.ai screens TB nationally.

IndiaAI Mission; PIB 2025

SOUTH KOREA

CLINICAL-AI EXPORTER

Naver HyperCLOVA X, LG EXAONE (open-weight), Samsung Gauss, backed by a national AI fund. Lunit + VUNO export FDA-cleared clinical AI worldwide.

MSIT; TechCrunch 2025

JAPAN

INNOVATION-FIRST

Sakana AI, the country's top unicorn; SoftBank owns ~11% of OpenAI. No-penalty AI law; AI-hospital pilots; Rapidus chasing a 2nm fab.

AI Promotion Act 2025

SOUTHEAST ASIA

THE BATTLEGROUND

SEA-LION switched from Meta's Llama to Alibaba's Qwen; Sahabat-AI (Indonesia), FPT (Vietnam). US and Chinese models compete for the next billion users.

TechNode 2025



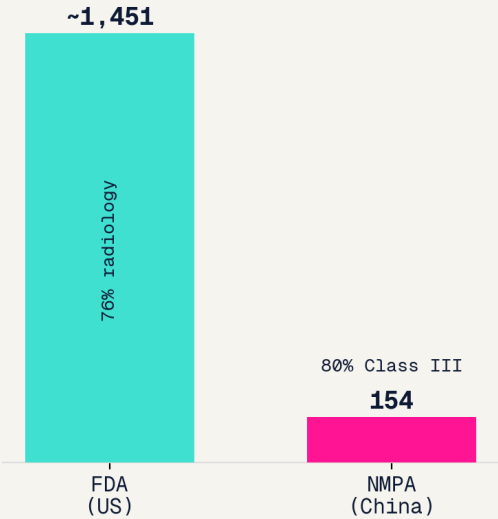
POLICY

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Regulating clinical AI

Clinical AI: the US approves the most discrete tools; China deploys at the most scale

Cumulative AI device approvals



but China DEPLOYS at

population scale

Ant Group's AQ app: 100M+ users, 5,000+ hospitals, ~1M doctors.
~261 hospitals ran DeepSeek-R1 in Q1 2025.

United States

Device-by-device 510(k) + PCCP. ~1,451 AI/ML devices cleared, 76% radiology; generative/LLM CDS still has no pathway.

Bedi et al., npj Digit Med 2025

Europe / EU

AI Act high-risk stacked on MDR/IVDR (double conformity); few dual-designated notified bodies; health-data space deferred to 2029.

EU AI Act + MDR/IVDR

Asia (China)

State-coordinated NMPA: 154 AI devices, ~80% the higher-risk Class III with mandatory clinical trials.

China NMPA AI-device review, JMIR 2026

Han et al. (Lancet Digit Health 2024): most clinical-AI RCTs remain few and early-stage.



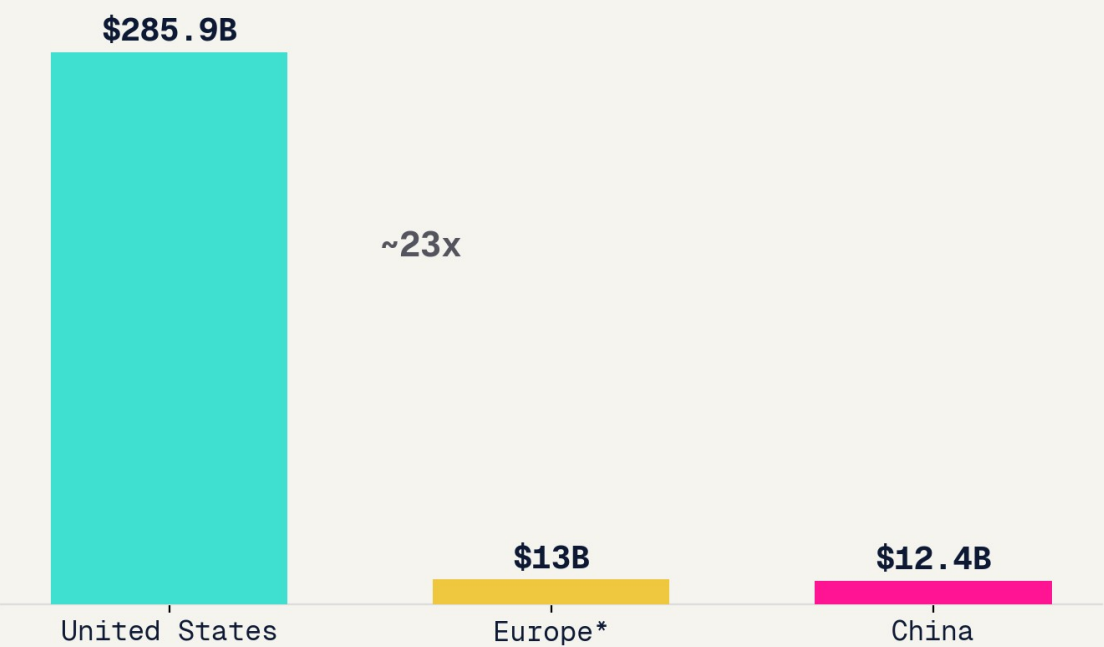
FUNDING

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Capital versus reimbursement

Private AI investment, 2025

Stanford AI Index (US, China). *Europe = est. EU+UK AI VC. The US captured ~83% of global private AI investment.



Stanford AI Index 2026. US ~83% of global private AI investment.

United States

Capital-rich, reimbursement-poor: mostly temporary CPT Category III codes + capped NTAP; few permanent Medicare pathways.

Health Tech Investment Act 2025

Europe / EU

National fast-tracks. Germany's DiGA reimburses ~60 apps (>1M prescriptions); France's PECAN copied it.

DiGA / BfArM

Asia (China)

State funding + volume-based procurement drive deployment without a Western-style reimbursement market.

NMPA / state procurement

Insurers also use AI to deny coverage.
Mello & Rose, JAMA Health Forum 2024

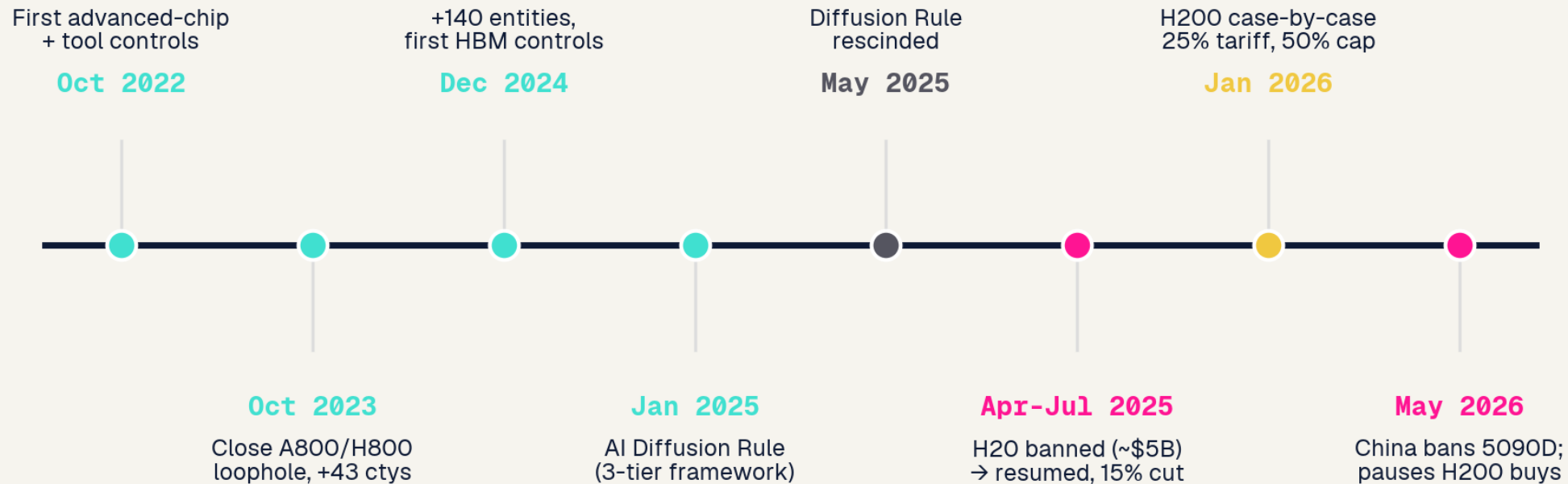


TECHNICAL DEVELOPMENTS

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The chip war reshapes the race

The chip war, 2022-2026: escalation, reversal, and China's pivot to domestic silicon



From the Oct 2022 controls to the Jan 2026 H200 case-by-case reversal, the GPU ceiling is the constraint that shapes every region's technical strategy.

Scale versus constraint: two paths to better models

UNITED STATES: SCALE

Answer the race with compute: Stargate (\$500B), ~\$700B hyperscaler capex in 2026, nuclear power deals. Frontier models stay closed. Abundant GPUs mean less pressure to economize.

Stargate; hyperscaler capex 2026

CHINA: CONSTRAINT

Capped on advanced GPUs by export controls, efficiency becomes the edge: mixture-of-experts, distillation, DeepSeek's ~\$6M training run, cheap inference. Necessity drives method innovation.

DeepSeek-V3 technical report, 2025

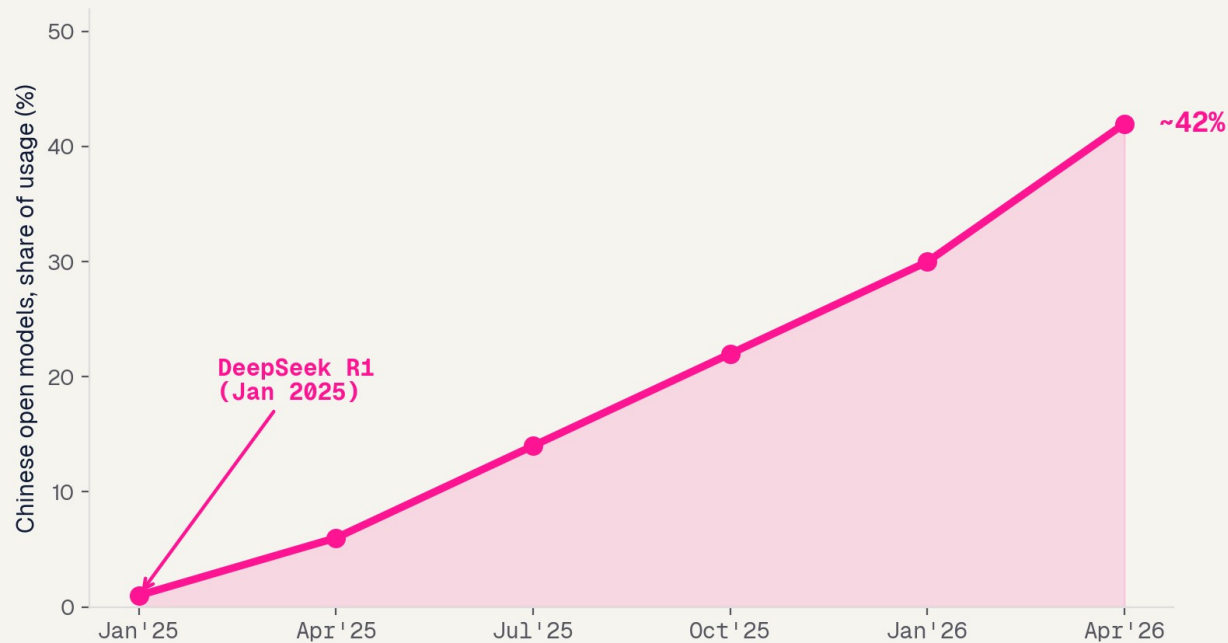
EUROPE: the upstream chokepoint. ASML's EUV monopoly is the gate the controls run through.

The provocative read: the GPU ceiling may be pushing China toward genuine methodology innovation, while the US mostly scales hardware. May 2026: Huawei's chairman publicly thanked the US for export controls, saying they 'supercharged' China's chip industry (Tom's Hardware; SCMP).

Open weights reach the bedside

China's open-weight surge

Open models (DeepSeek, Qwen) rose from ~1% to ~40%+ of OpenRouter usage in a year; HF downloads China 17.1% > US 15.9%



Chinese open models, share of usage on a major model router.

The DeepSeek moment

An open, frontier-quality reasoning model at low cost reset the board in Jan 2025; ~\$600B wiped off NVIDIA in a day.

NVIDIA -17% selloff, Jan 2025

Into hospitals, fast

~261 Chinese hospitals locally deployed DeepSeek-R1 in Q1 2025 (84% tertiary), outpacing regulation.

DeepSeek-R1 in 261 hospitals, Nat Med 2025

The Western void

FDA's Jan 2026 CDS guidance is silent on generative/LLM clinical AI; the EU defaults it to high-risk.

FDA CDS guidance, Jan 2026

RCT caution: an LLM did not significantly improve physician diagnosis. Goh et al., JAMA Netw Open 2024

The frontier standing: who actually leads

UNITED STATES

Holds the absolute frontier

Claude Opus 4.8, GPT-5.5, Gemini 3.1 lead the top tier; 50 notable models in 2025 vs China's 30.

artificialanalysis.ai; Stanford AI Index 2026

ASIA (CHINA)

Fast follower, cheap + open

Within ~2.7% on Arena, ~7-month lag. DeepSeek V4 (Apr 2026) near-frontier at ~1/10 the cost; dominates open weights.

Epoch AI; Stanford AI Index 2026

EUROPE / EU

Thin at the frontier

Mistral is the lone challenger; DeepMind sits under US (Google) ownership.

Draghi report, 2024

The frontier crown is still American. China's edge is open weights and cost, not the top tier, and the proprietary lead may even be widening.

Evaluating clinical AI: can we even measure it?

HEALTHBENCH

Physician-built rubrics (262 doctors, 5,000 conversations) grade how well LLMs answer real health questions, the first serious clinical-LLM benchmark.

Arora et al., HealthBench, 2025

HEALTHADMINBENCH

Tests computer-use agents on real healthcare administration tasks, prior auth, billing, scheduling, the unglamorous work that dominates cost.

Bedi et al., HealthAdminBench, 2026

The limit: benchmarks measure what a model can say, graded against expert opinion, not whether it improves patient outcomes. We are getting good at testing capability, not benefit.



LANDSCAPE

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Who deploys clinical AI at scale

UNITED STATES

- ~1,451 AI/ML devices cleared; 76% radiology
- Reimbursement-gated: temporary CPT III codes
- Slowest path from clearance to bedside

Bedi et al., npj Digit Med 2025

EUROPE / EU

- AI Act high-risk + MDR/IVDR double burden
- Nationwide AI mammography deployed (Germany)
- DiGA the working reimbursement route

Eisemann et al., Nat Med 2025

ASIA (CHINA-LED)

- Ant Group AQ app: 100M+ users, 5,000+ hospitals
- ~261 hospitals ran DeepSeek-R1 in Q1 2025
- Korea exports: Lunit FDA-cleared, ~1M mammograms/yr

DeepSeek-R1 in 261 hospitals, Nat Med 2025

Approvals and deployment are different axes: the US leads on count, Asia on scale.

Real-world clinical AI, by region

UNITED STATES

Commercial scale, narrow tools

Aidoc: 1,600+ centers, 100M+ cases. Viz.ai stroke: 1,800 hospitals. FDA-cleared, radiology and acute triage.

Aidoc; Viz.ai, 2025-26

EUROPE / EU

The autonomy milestone

Oxipit ChestLink: world-first FULLY AUTONOMOUS radiology AI (CE Class IIb), auto-reports ~40% of normal chest X-rays, a clearance the FDA has not granted.

Oxipit / Sectra, 2026

ASIA (CHINA)

Population-scale screening

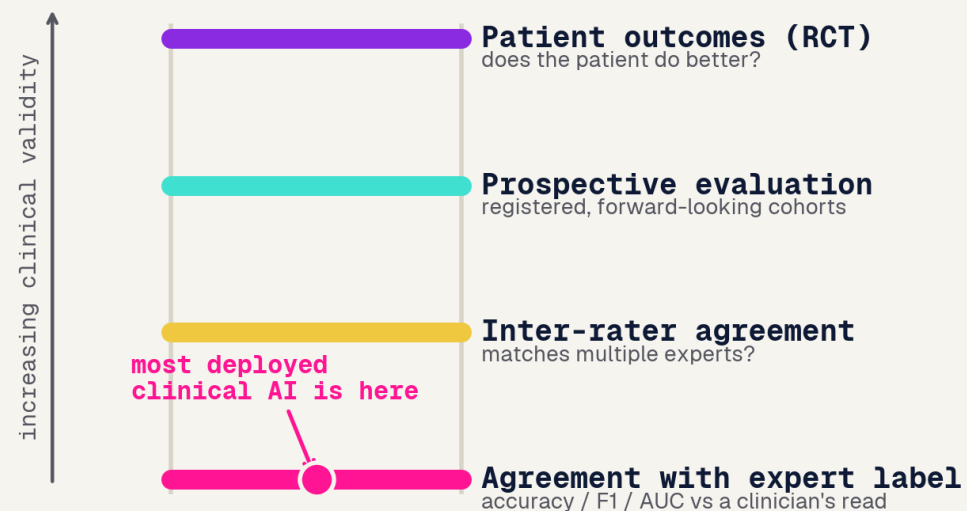
Infervision ~20k lung scans/day; Tencent Miying in 100+ hospitals; Ant AQ 100M users; DeepSeek-R1 in 261 hospitals.

Nature Medicine 2025; co. data

Real systems, same split: the US scales narrow tools, Europe pioneers autonomy under stricter rules, China deploys at population scale.

The ROI reality

The validation gap: agreement is not benefit



Agreement with a label is not benefit to a patient.

95%

of enterprise gen-AI pilots showed no measurable P&L impact (MIT NANDA 2025)

Who closes the validation gap?

UNITED STATES

Reimbursement scrutiny forces some evidence, but approval is still label-based.

EUROPE / EU

Demands evidence ex-ante (AI Act + MDR), yet the health-data space is deferred to 2029.

ASIA (CHINA)

Deploys before outcomes are proven, the fastest rollout, the widest validation gap.

Han, Lancet Digit Health 2024 - Challa et al. (MIT NANDA) 2025

Clinician and AI: help or hazard?

THE PROMISE

AI augments the clinician, extending scarce expertise and catching what a tired human misses. The pitch behind every deployment.

THE RCT REALITY

Giving physicians an LLM did not significantly improve diagnostic reasoning vs usual resources, the model alone scored well; doctors underused it.

Goh et al., JAMA Netw Open 2024

THE HAZARD

Over-reliance breeds automation bias and deskilling, e.g. endoscopists detected fewer adenomas after routine AI use.

Budzyn et al., Lancet Gastro 2025

Success hinges on the human-AI interaction, not accuracy. And the evidence is lopsided: the US and EU run these RCTs and automation-bias studies; China's population-scale deployment has outpaced interaction research.

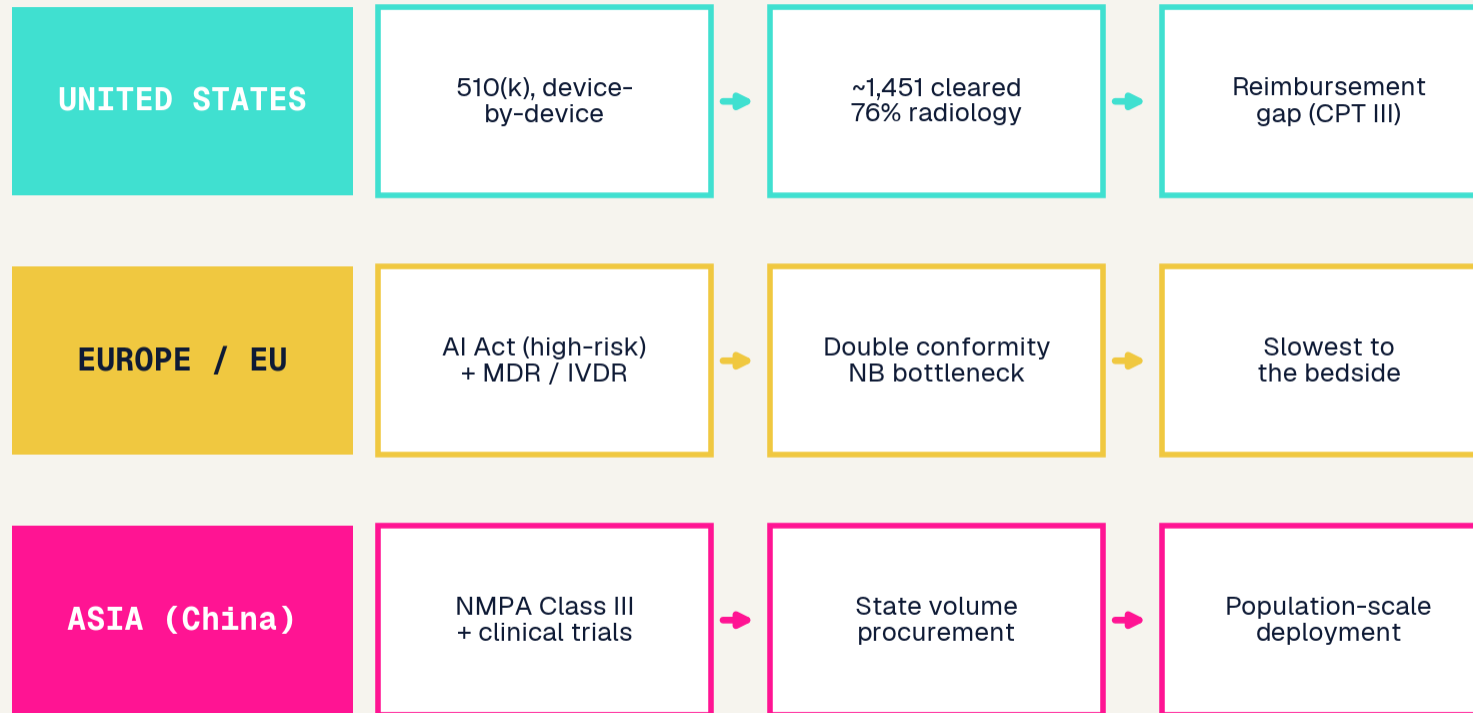


SYNTHESIS

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Regulate, build, deploy, in medicine

How clinical AI reaches patients: three pathways



The bottleneck on deployed clinical AI is institutional, regulation, payment, and coordination, not model quality.

Takeaways and outlook

01

The bottleneck on deployed clinical AI is institutional, regulation, payment, coordination, not model quality.

02

Same three governing systems define clinical AI: the US approves, Europe regulates, Asia deploys.

03

Open weights reach the bedside fastest under state coordination; the US and EU still lack a generative-clinical-AI pathway.

04

Measure what matters: shift clinical-AI evaluation from agreement-with-labels to patient outcomes.

Key references

#	Reference	Venue / year
1	Han et al. - RCTs evaluating AI in clinical practice	Lancet Digital Health 2024
2	Goh et al. - LLM influence on diagnostic reasoning (RCT)	JAMA Network Open 2024
3	Eisemann et al. - Nationwide AI in mammography screening	Nature Medicine 2025
4	Bedi et al. - FDA AI/ML-enabled device landscape	npj Digital Medicine 2025
5	China NMPA AI-device approvals (review)	JMIR Med Informatics 2026
6	DeepSeek-R1 deployed across Chinese hospitals	Nature Medicine 2025
7	Mello & Rose - AI tools and insurance coverage denials	JAMA Health Forum 2024
8	Maslej et al. - Stanford AI Index Report 2026	Stanford HAI 2026
9	Challa et al. (MIT NANDA) - State of AI in Business	MIT NANDA 2025
10	Arora et al. - HealthBench	arXiv 2025
11	DeepSeek-AI - DeepSeek-V4 technical report	arXiv 2026
12	Epoch AI - US-China model capability gap	Epoch AI 2026

Framing: Lee, AI Superpowers (2018) - Schmidt et al., The Age of AI (2021) - Draghi, EU Competitiveness Report (2024).

Full source list with URLs in the companion research.md.

Thank You

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夢想